

Suricata IDS Lab

Detection, Validation & Controlled Exploitation Project

Ubuntu Server	Suricata IDS — 192.168.56.20
Kali Linux	Testing / reconnaissance — 192.168.56.10
Metasploitable 2	Vulnerable target — 192.168.56.30

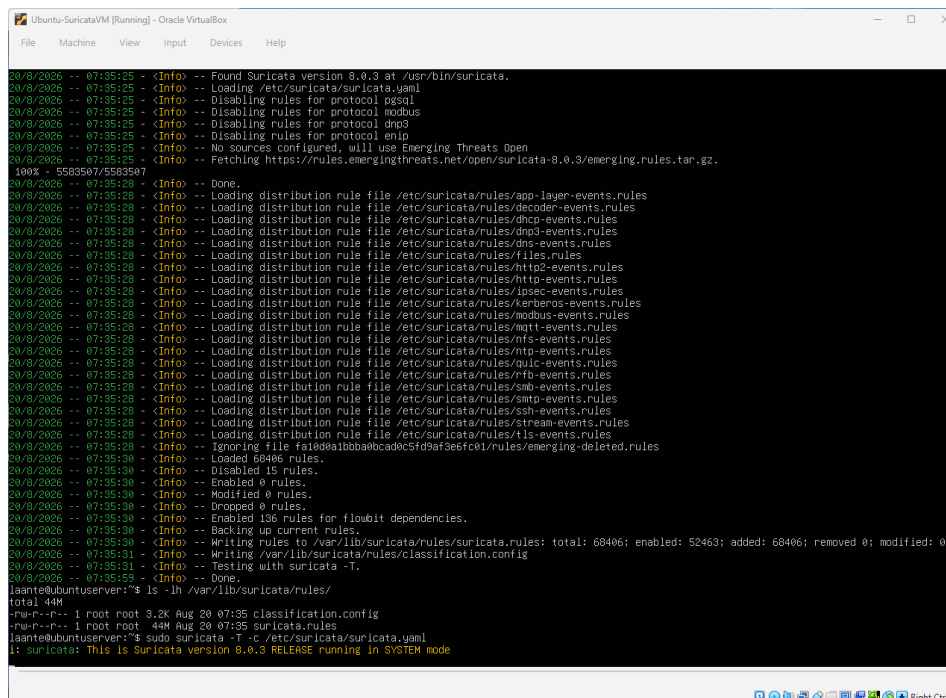
Objective: Build and validate an isolated Suricata IDS lab, generate controlled network activity, detect it with custom and managed rules, investigate EVE JSON alerts, and validate detection during controlled exploitation of Metasploitable 2.

1. Lab Architecture

The lab used three VirtualBox systems. Ubuntu Server acted as the IDS sensor, Kali generated reconnaissance and application traffic, and Metasploitable 2 provided an intentionally vulnerable target. Suricata monitored the 192.168.56.0/24 lab network through enp0s3.

2. Suricata Deployment & Configuration

Suricata 8.0.3 was installed, managed rules were loaded, and the configuration was validated with `suricata -T`. The configuration successfully loaded the rule set and custom `local.rules`. Suricata was then started and verified as an active systemd service.



```
20/8/2026 -- 07:35:25 - <Info> -- Found Suricata version 8.0.3 at /usr/bin/suricata.
20/8/2026 -- 07:35:25 - <Info> -- Loading /etc/suricata/suricata.yaml
20/8/2026 -- 07:35:25 - <Info> -- Disabling rules for protocol psql
20/8/2026 -- 07:35:25 - <Info> -- Disabling rules for protocol modbus
20/8/2026 -- 07:35:25 - <Info> -- Disabling rules for protocol dnss
20/8/2026 -- 07:35:25 - <Info> -- Disabling rules for protocol enip
20/8/2026 -- 07:35:25 - <Info> -- No sources configured, will use Emerging Threats Open
20/8/2026 -- 07:35:25 - <Info> -- Fetching https://rules.emergingthreats.net/open/suricata-8.0.3/emerging.rules.tar.gz.
100% - 5583597/5583597
20/8/2026 -- 07:35:28 - <Info> -- Done.
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/app-layer-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/decoder-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/dhcp-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/dnp3-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/dns-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/files.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/http2-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/http-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/isec-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/kerberos-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/modbus-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/mqtt-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/nfs-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/ntp-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/quic-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/rfb-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/smb-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/smb-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/ssh-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/stream-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Loading distribution rule file /etc/suricata/rules/tls-events.rules
20/8/2026 -- 07:35:28 - <Info> -- Ignoring file fa10da1bbaabca0c5fd9af3e6fcd1/rules/emerging-deleted.rules
20/8/2026 -- 07:35:30 - <Info> -- Loaded 60406 rules.
20/8/2026 -- 07:35:30 - <Info> -- Disabled 15 rules.
20/8/2026 -- 07:35:30 - <Info> -- Enabled 0 rules.
20/8/2026 -- 07:35:30 - <Info> -- Modified 0 rules.
20/8/2026 -- 07:35:30 - <Info> -- Dropped 0 rules.
20/8/2026 -- 07:35:30 - <Info> -- Enabled 136 rules for flowbit dependencies.
20/8/2026 -- 07:35:30 - <Info> -- Backing up current rules.
20/8/2026 -- 07:35:30 - <Info> -- Writing rules to /var/lib/suricata/rules/suricata.rules: total: 60406; enabled: 52463; added: 60406; removed 0; modified: 0
20/8/2026 -- 07:35:31 - <Info> -- Writing /var/lib/suricata/rules/classification.config
20/8/2026 -- 07:35:31 - <Info> -- Testing with suricata -T.
20/8/2026 -- 07:35:35 - <Info> -- Done.
laante@ubuntu-server:~$ ls -ln /var/lib/suricata/rules/
total 44M
-rw-r--r-- 1 root root 3.2K Aug 20 07:35 classification.config
-rw-r--r-- 1 root root 44M Aug 20 07:35 suricata.rules
laante@ubuntu-server:~$ sudo suricata -T -c /etc/suricata/suricata.yaml
!: suricata: This is Suricata version 8.0.3 RELEASE running in SYSTEM mode
```

Figure 1 — Suricata configuration validation and rule loading.

```
Ubuntu-SuricataVM [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

- interface: enp0s3
# Number of receive threads. "auto" uses the number of cores
# threads: auto
# Default clusterId. AF_PACKET will load balance packets based on flow.
# clusterId: 99
# Default AF_PACKET cluster type. AF_PACKET can load balance per flow or per hash.
# This is only supported for Linux kernel > 3.1
# possible value are:
# * cluster_flow: all packets of a given flow are sent to the same socket
# * cluster_cpu: all packets treated in kernel by a CPU are sent to the same socket
# * cluster_qm: all packets linked by network card to a RSS queue are sent to the same
# socket. Requires at least Linux 3.14.
# * cluster_bpfp: eBPF file load balancing. See doc/userguide/capture-hardware/ebpf-xdp.rst for
# more info.
# Recommended modes are cluster_flow on most boxes and cluster_cpu on cluster_qm on system
# with capture card using RSS (requires cpu affinity tuning and system IRQ tuning)
# cluster_rollover has been deprecated; if used, it'll be replaced with cluster_flow.
cluster-type: cluster_flow
# In some fragmentation cases, the hash can not be computed, if "defrag" is set
# to yes, the kernel will do the needed defragmentation before sending the packets.
defrag: yes
# Lock memory map to avoid it being swapped. Be careful that over
# subscribing could lock your system

laante@ubuntu:~$ sudo suricata -T -c /etc/suricata/suricata.yaml
i: suricata: This is Suricata version 8.0.3 RELEASE running in SYSTEM mode
i: rule-base: Rule group caching - loaded: 117 newly cached: 0 total cacheable: 117
i: suricata: Configuration provided was successfully loaded. Exiting.
laante@ubuntu:~$ sudo systemctl restart suricata
laante@ubuntu:~$ sudo systemctl status suricata --no-pager
* suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-08-20 07:58:47 UTC; 27s ago
     Invocation: ae7a87ad4493477b9ab9ae910662b7bb
       Docs: man:suricata(8)
            man:suricatasc(8)
            https://suricata.io/documentation/
    Main PID: 1030 (Suricata-Main)
      Tasks: 6 (limit: 1728)
     Memory: 416.3M (peak: 416.3M)
        CPU: 18.236s
    CGroup: /system.slice/suricata.service
            └─1030 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml -pidfile /var/run/suricata/suricata.pid

Aug 20 07:58:29 ubuntu:systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Aug 20 07:58:32 ubuntu:systemd[1]: suricata: This is Suricata version 8.0.3 RELEASE running in SYSTEM mode
Aug 20 07:58:47 ubuntu:systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
laante@ubuntu:~$ sudo tail -f /var/log/suricata/fast.log
```

Figure 2 — Suricata service running under systemd.

3. Detection Engineering

Custom IDS signatures were created for specific lab behaviors. Alerts were validated directly in /var/log/suricata/eve.json, establishing a complete chain from generated traffic to structured IDS telemetry.

Detection	Observed activity	Result
ICMP	Kali ping to sensor	IDS LAB – ICMP Ping Detected
SSH	SSH probe activity	IDS LAB – Nmap SSH Probe Detected
HTTP	HTTP request to /admin	IDS LAB – Suspicious HTTP Admin Request
DNS	DNS query traffic	IDS LAB – DNS Query Detected
TCP SYN	Nmap service probing	IDS LAB – TCP SYN Probe Detected

4. Live Detection Evidence

Kali-generated traffic was observed in real time by Suricata. The evidence shows source and destination addresses, protocols, ports, signature IDs and alert names in the EVE JSON stream.

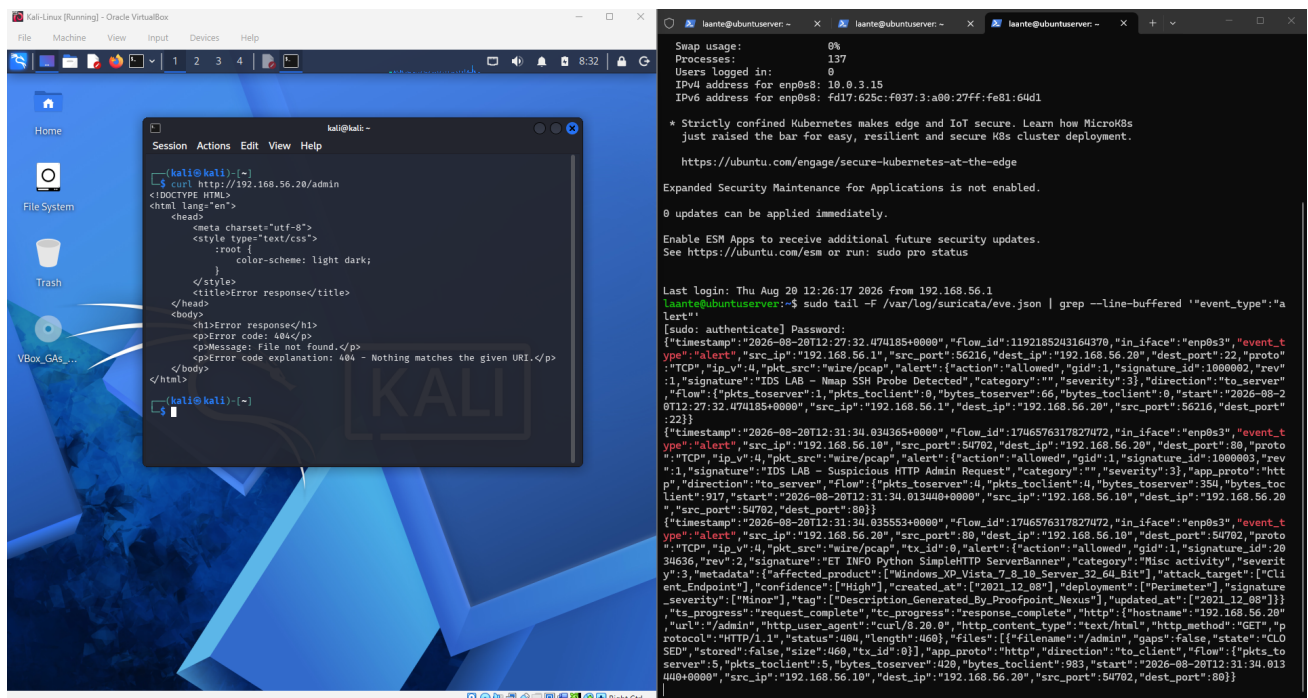


Figure 3 — HTTP request and Suricata HTTP/SSH detections.

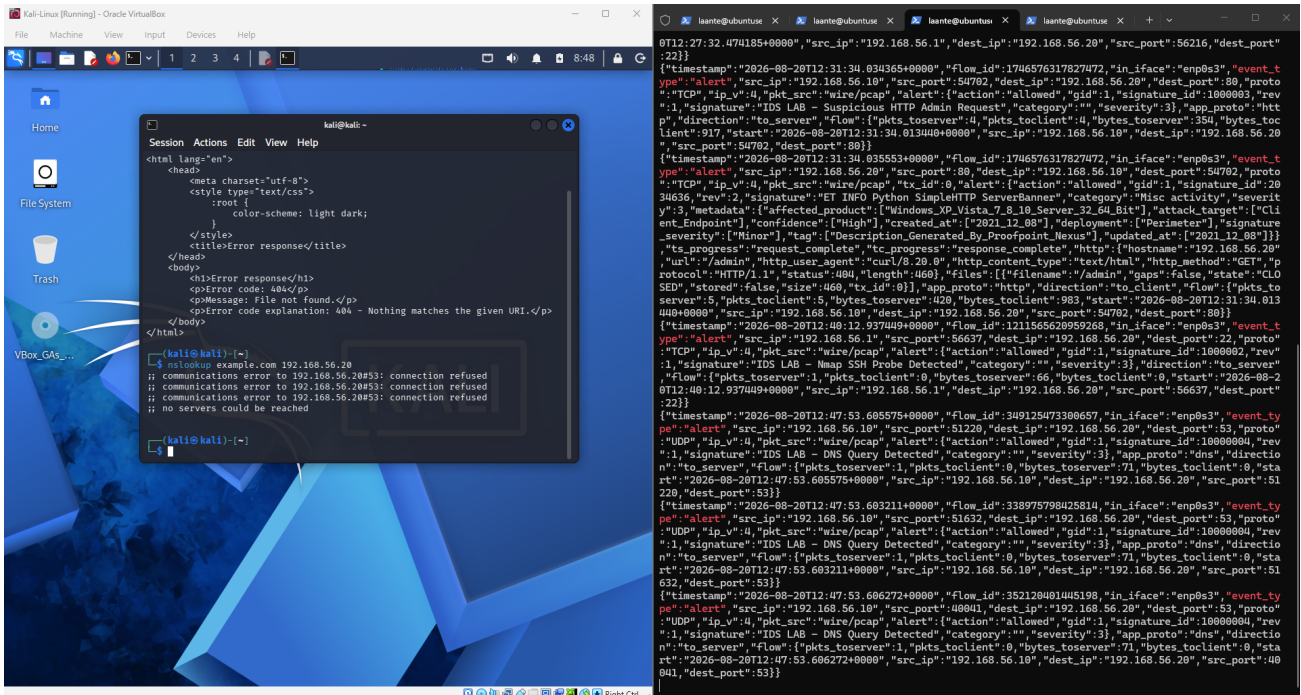


Figure 4 — DNS traffic and Suricata DNS Query detection.

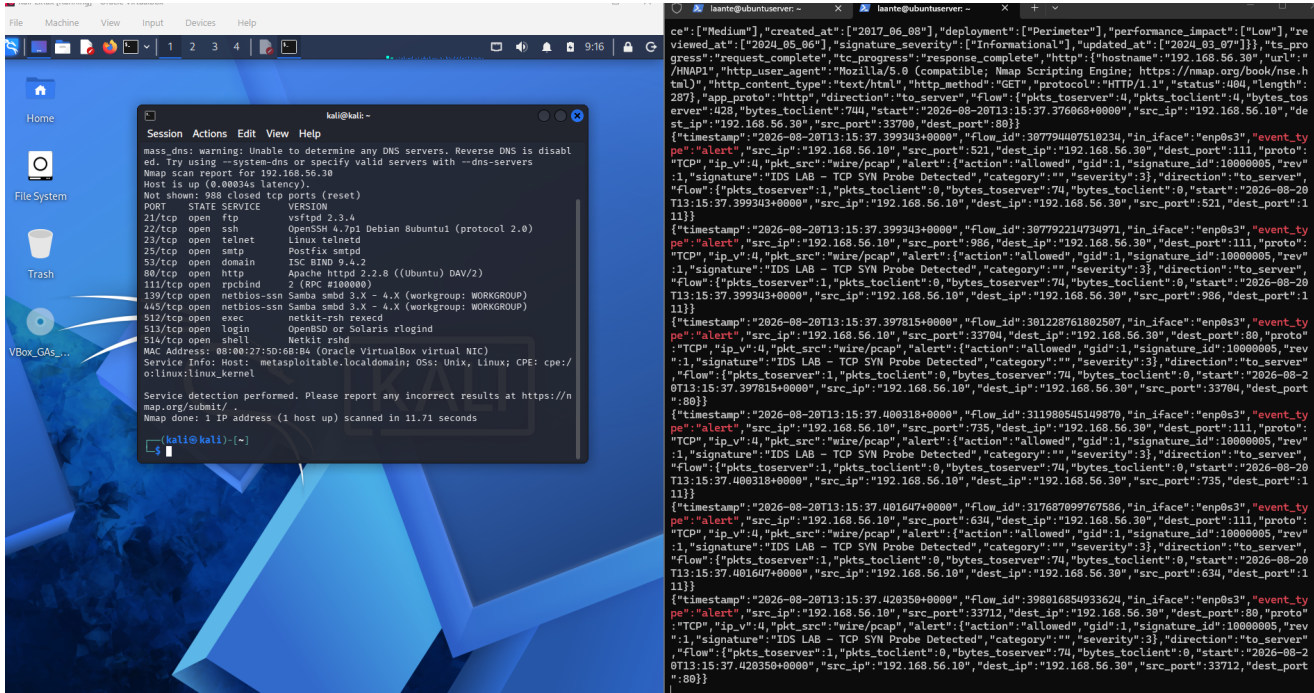


Figure 5 — Nmap reconnaissance and Suricata TCP SYN detections.

5. Metasploitable Reconnaissance

Metasploitable 2 was added as the intentionally vulnerable target at 192.168.56.30. Nmap identified a broad attack surface including FTP (21), SSH (22), Telnet (23), SMTP (25), DNS (53), HTTP (80), SMB/NetBIOS (139/445), and other legacy services. Suricata simultaneously generated TCP SYN probe alerts for the reconnaissance traffic.

6. Controlled Exploitation

For final validation, the intentionally vulnerable vsftpd 2.3.4 service was tested with Metasploit's exploit/unix/ftp/vsftpd_234_backdoor module. RHOSTS was 192.168.56.30, RPORT 21, LHOST 192.168.56.10 and LPORT 4444. The exploit produced a Meterpreter session, while Suricata recorded the associated network activity.

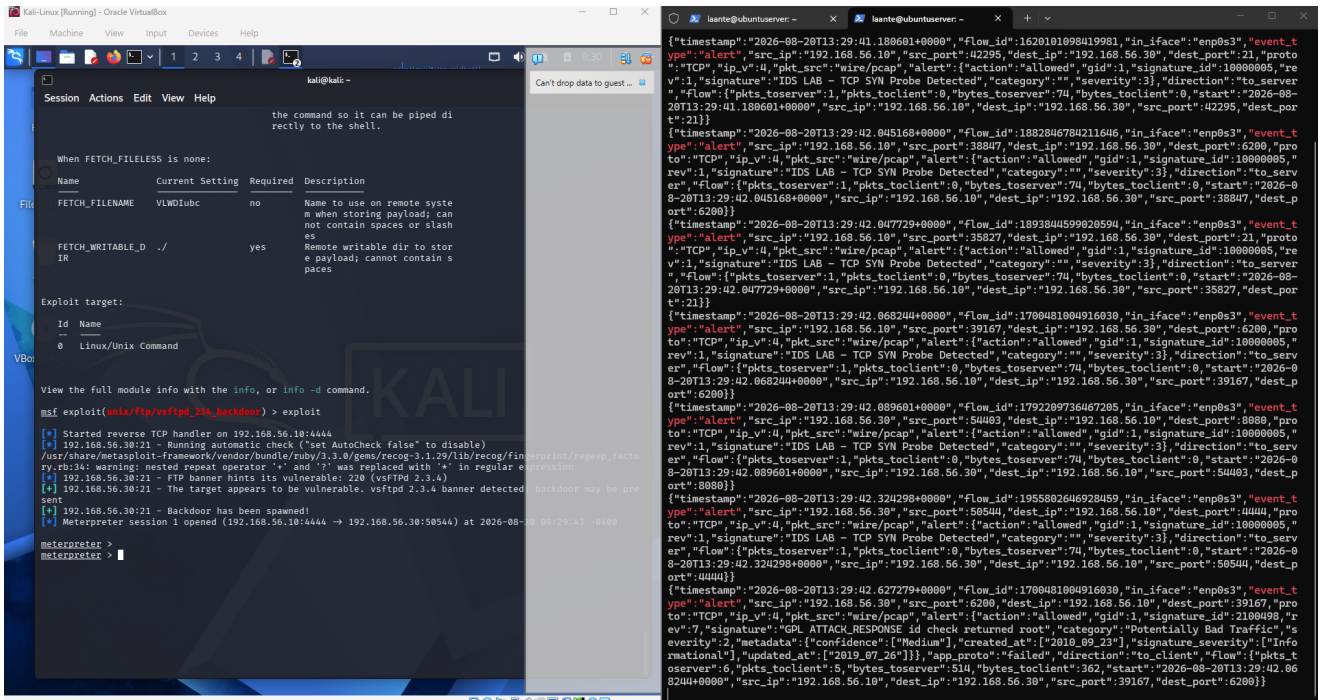


Figure 6 — Controlled vsftpd exploitation with Metasploit and corresponding Suricata telemetry.

7. Final Results

Validation	Status
Suricata configuration	Successful
Managed and custom rules	Loaded successfully
EVE JSON logging	Operational
ICMP / SSH / HTTP detection	Verified
DNS detection	Verified
Nmap TCP SYN detection	Verified
Metasploitable reconnaissance	Verified
Controlled vsftpd exploitation	Successful in isolated lab
Attack-response telemetry	Observed in Suricata logs

8. SOC-Relevant Skills Demonstrated

- Linux command-line administration and troubleshooting
- Suricata IDS installation, configuration and service management
- Network interface and packet-capture configuration
- Custom IDS signature creation and validation
- EVE JSON alert analysis and filtering
- Nmap reconnaissance and service enumeration
- Network traffic generation with Kali Linux
- Detection validation against an intentionally vulnerable host
- Correlation of source, destination, protocol, ports and signatures
- Controlled exploitation testing with Metasploit

9. Conclusion

The project demonstrates a complete introductory IDS workflow: deploy the sensor, configure packet capture and rules, generate controlled traffic, validate alerts, investigate structured telemetry, and test detection during a controlled exploitation scenario. The screenshots provide direct evidence of the work performed in the isolated VirtualBox environment.